

BLUESTOCK FINTECH

Mutual Fund Analytics Platform — Capstone Project

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1. Executive Summary

This project builds an end-to-end Mutual Fund Analytics Platform using publicly available data from AMFI India. The platform covers data ingestion, cleaning, EDA, performance metrics and an interactive Power BI dashboard. The Indian MF industry manages over Rs.81 lakh crore in AUM across 1908 schemes and 26.12 crore investor folios as of December 2025. Monthly SIP inflows hit all-time high of Rs.31,002 crore in December 2025.

2. Datasets Used

10 datasets were used in this project. 01_fund_master has master list of 40 mutual fund schemes. 02_nav_history has daily NAV from 2022 to 2026. 03_aum_by_fund_house has quarterly AUM for 10 fund houses. 04_monthly_sip_inflows has monthly SIP data. 05_category_inflows has category wise net inflows. 06_industry_folio_count has total folio growth. 07_scheme_performance has Sharpe, Alpha, Beta metrics. 08_investor_transactions has 32778 transactions. 09_portfolio_holdings has top equity holdings. 10_benchmark_indices has Nifty 50 and Nifty 100 data.

3. ETL Pipeline

EXTRACT: Loaded all 10 CSV files using Python Pandas. Fetched live NAV data from mfapi.in REST API for 5 schemes.

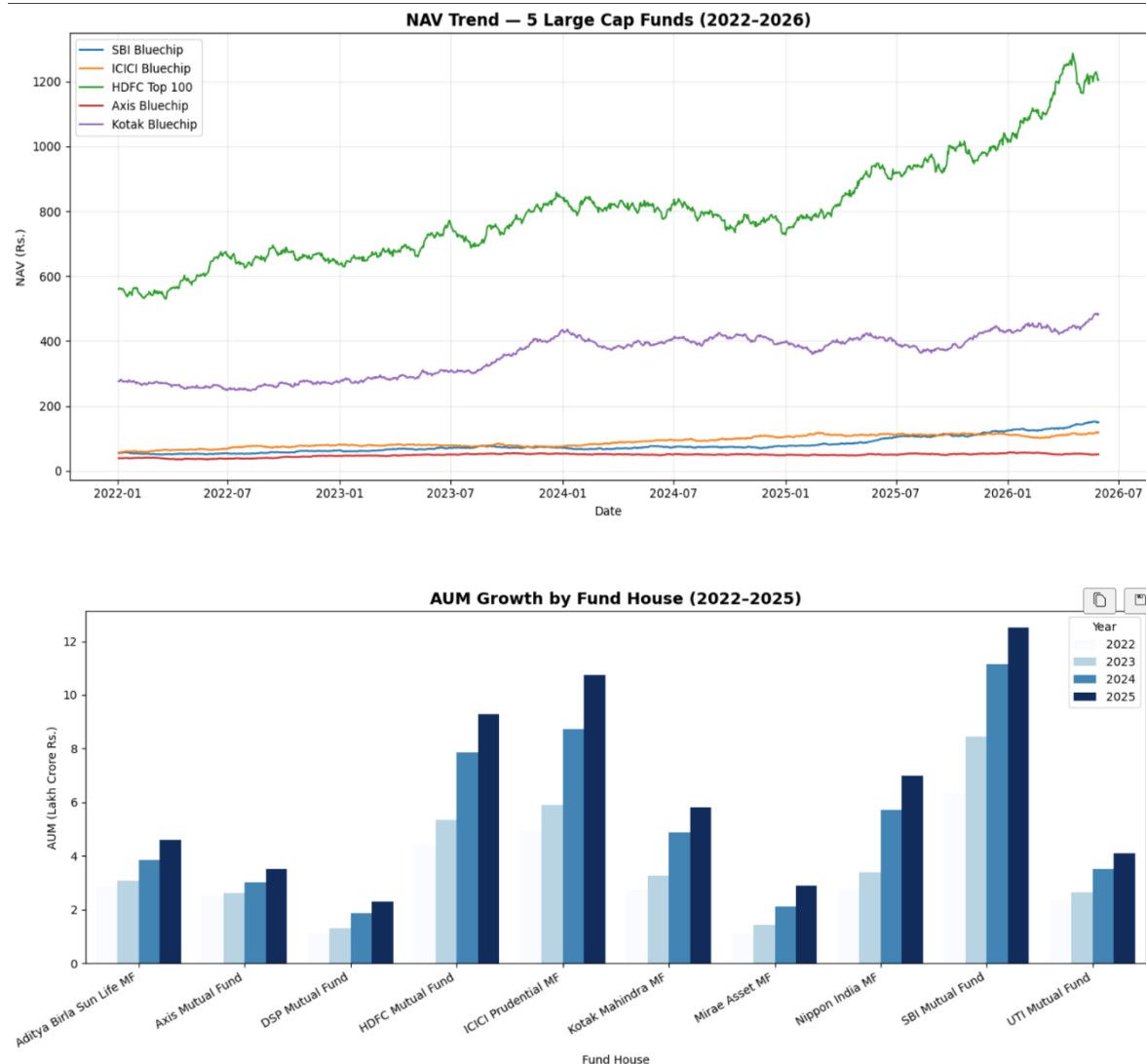
TRANSFORM: Parsed all date columns to datetime. Removed duplicate rows. Forward filled missing NAV values for weekends and holidays. Standardised transaction types. Validated AMFI codes across all datasets.

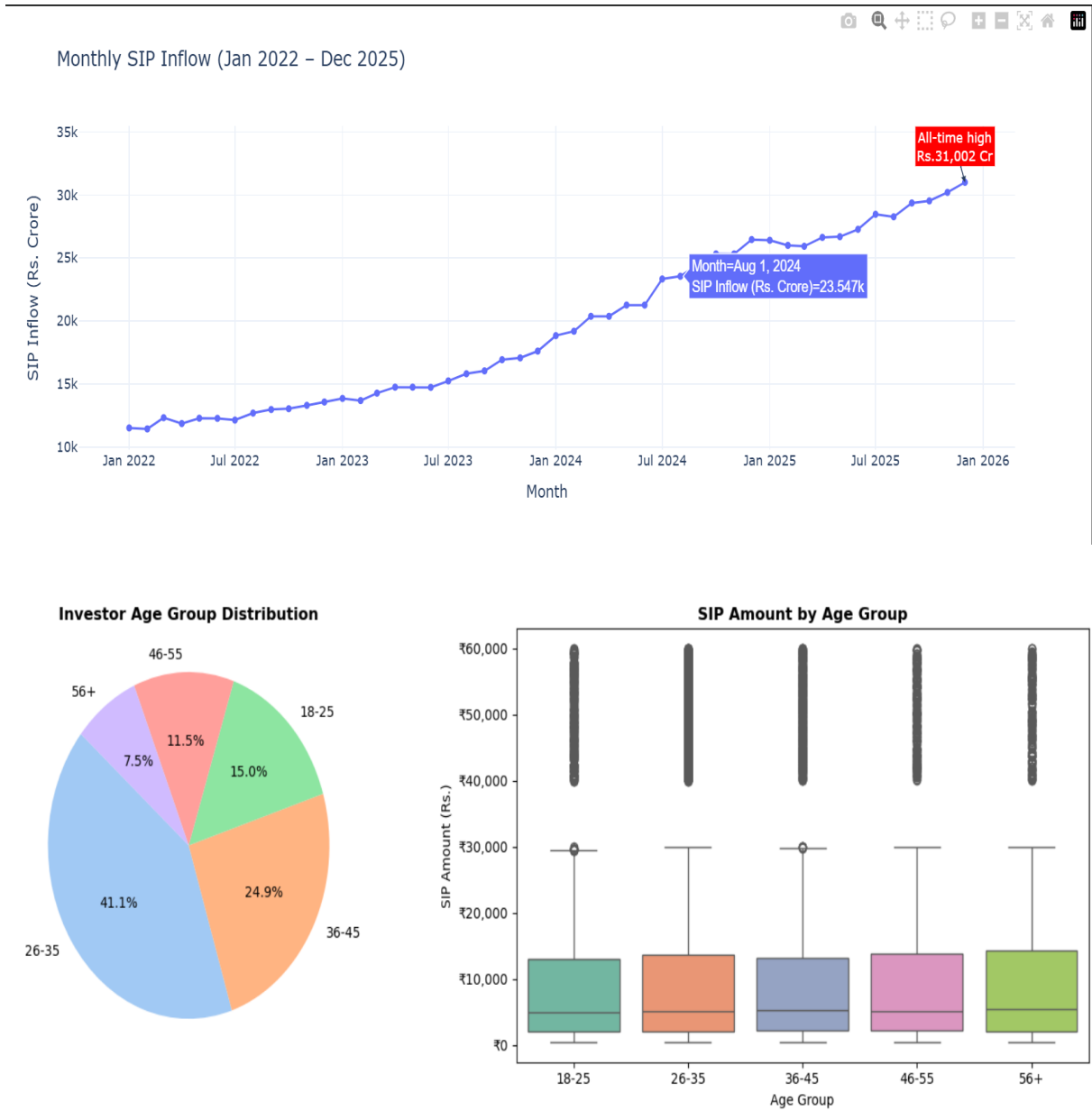
LOAD: Saved all cleaned files to processed folder. Loaded all 8 tables into SQLite database. Ran 10 SQL queries to verify data integrity.

4. EDA Key Findings

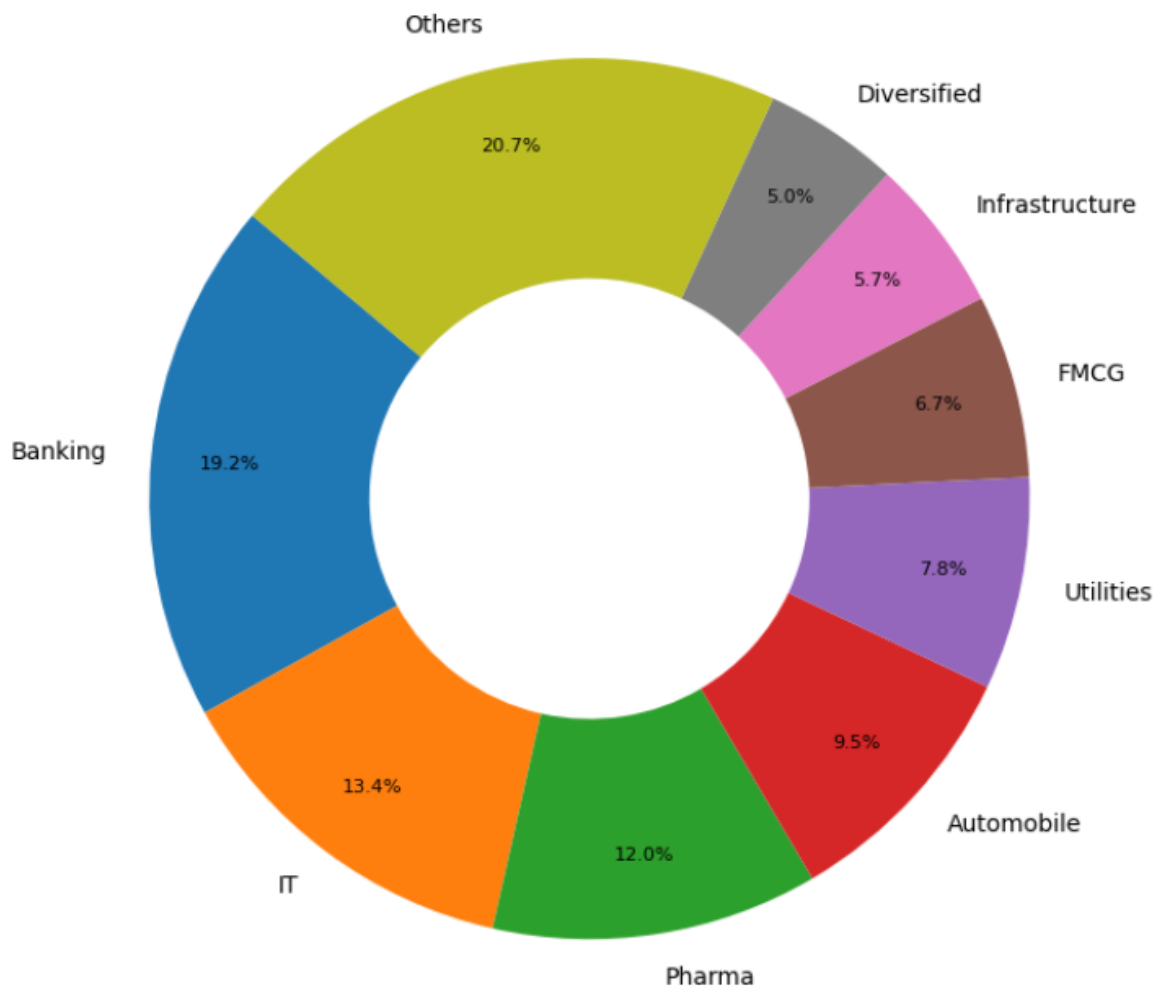
1. SBI MF dominates with Rs.12.5 lakh crore AUM, nearly 2x ICICI Pru.

2. SIP inflows grew from Rs.11,517 crore in Jan 2022 to approximately Rs.31,000 crore in Dec 2025.
3. Total folios doubled from 13.26 crore to over 26 crore folios in 4 years.
4. 26-35 age group is the largest SIP investor segment.
5. Maharashtra, Karnataka and Delhi lead in SIP investment volumes.
6. Financial Services and IT account for over 40% of equity portfolios.
7. Large cap funds show high correlation above 0.85 with each other.
8. T30 cities contribute over 70% of total investment amount.





Sector Allocation Across Equity Fund Portfolios



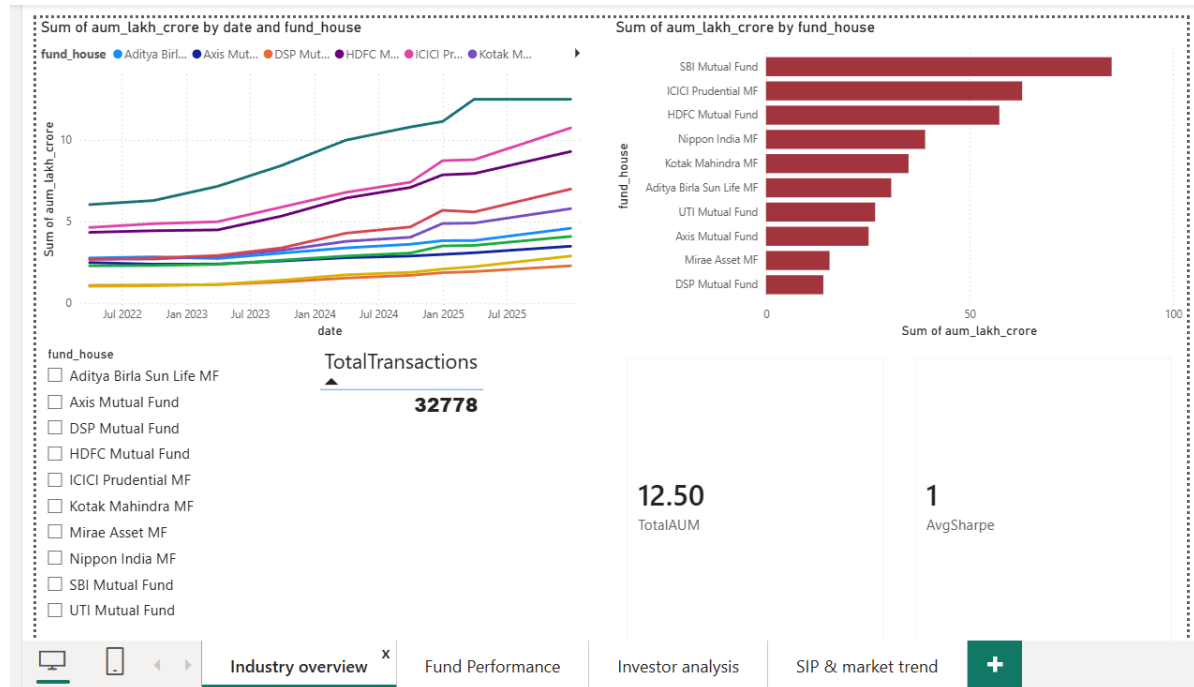
5. Performance Analysis

Daily returns computed as $\text{NAV}_t / \text{NAV}_{t-1} - 1$. CAGR computed for 1yr, 3yr and 5yr periods. Sharpe Ratio calculated using RBI Repo Rate of 6.5% as risk free rate. Alpha and Beta computed using OLS regression vs Nifty 100 benchmark. Maximum Drawdown computed as worst peak to trough decline.

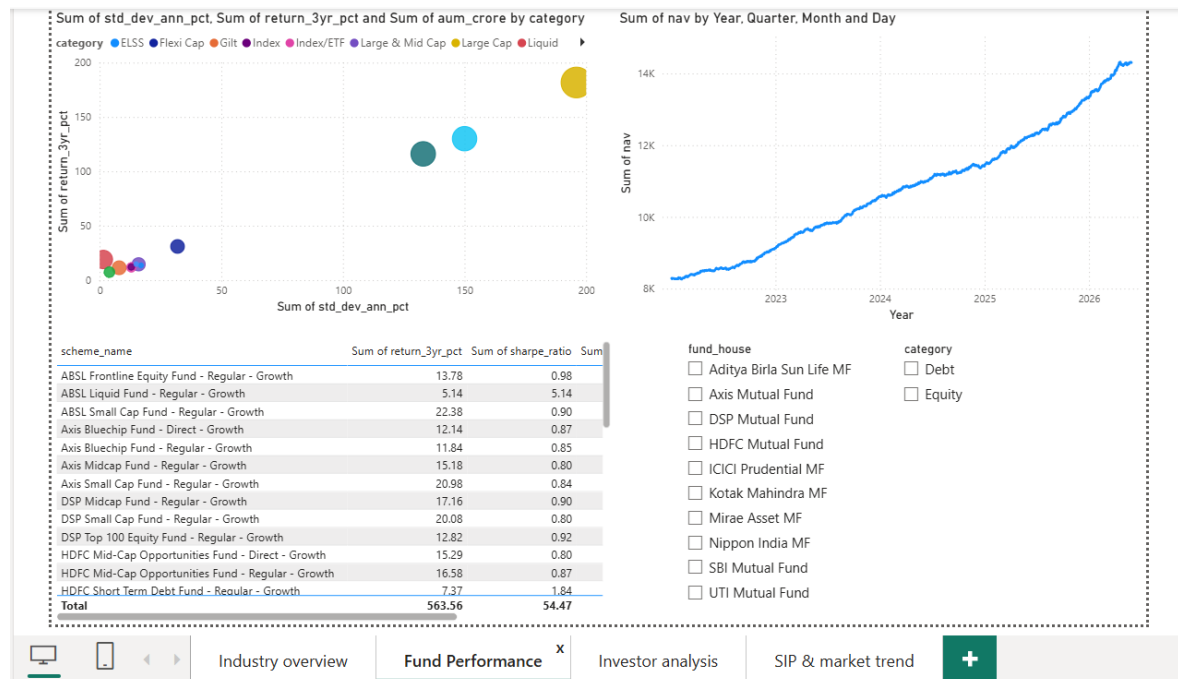
Fund Scorecard: Composite Score = 30% x 3yr Return Rank + 25% x Sharpe Rank + 20% x Alpha Rank + 15% x Expense Ratio Rank + 10% x Max Drawdown Rank.

6.Power BI Dashboard

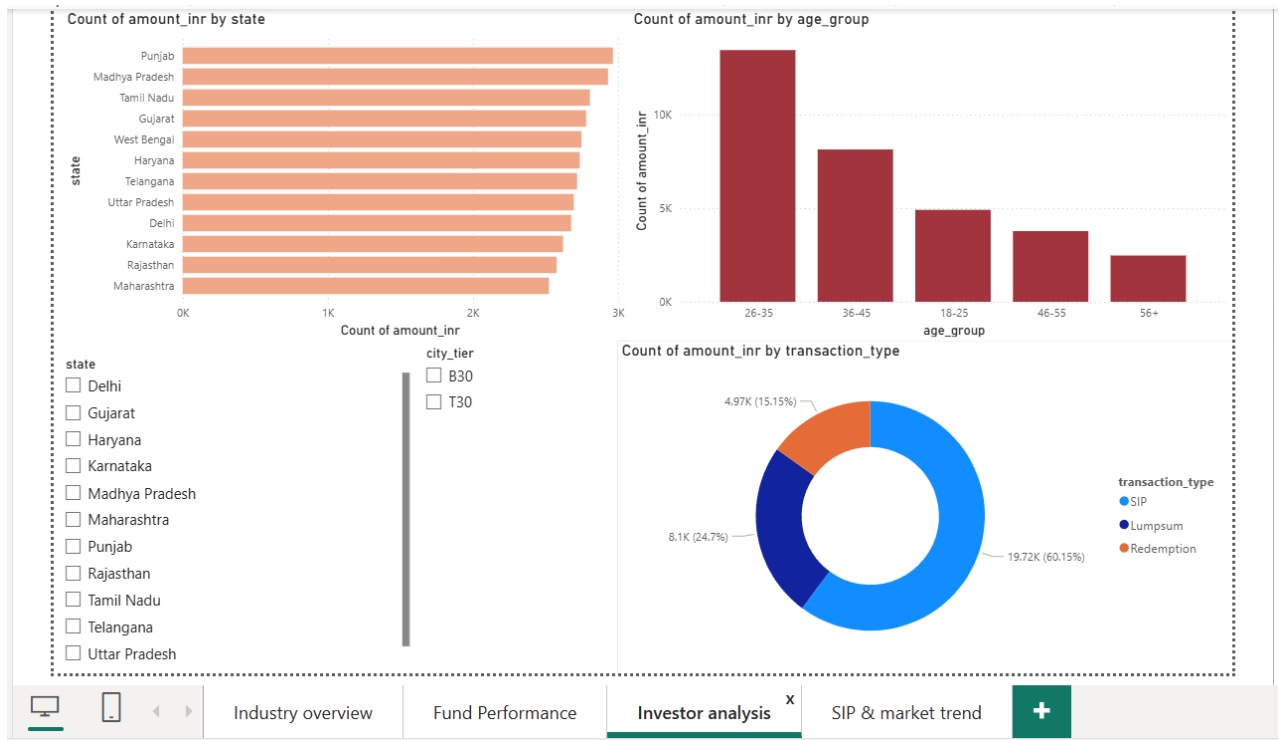
PAGE 1 - Industry Overview: KPI cards for Total AUM, SIP Inflow and Total Transactions. Line chart for AUM trend 2022-2025. Bar chart for AUM by fund house.



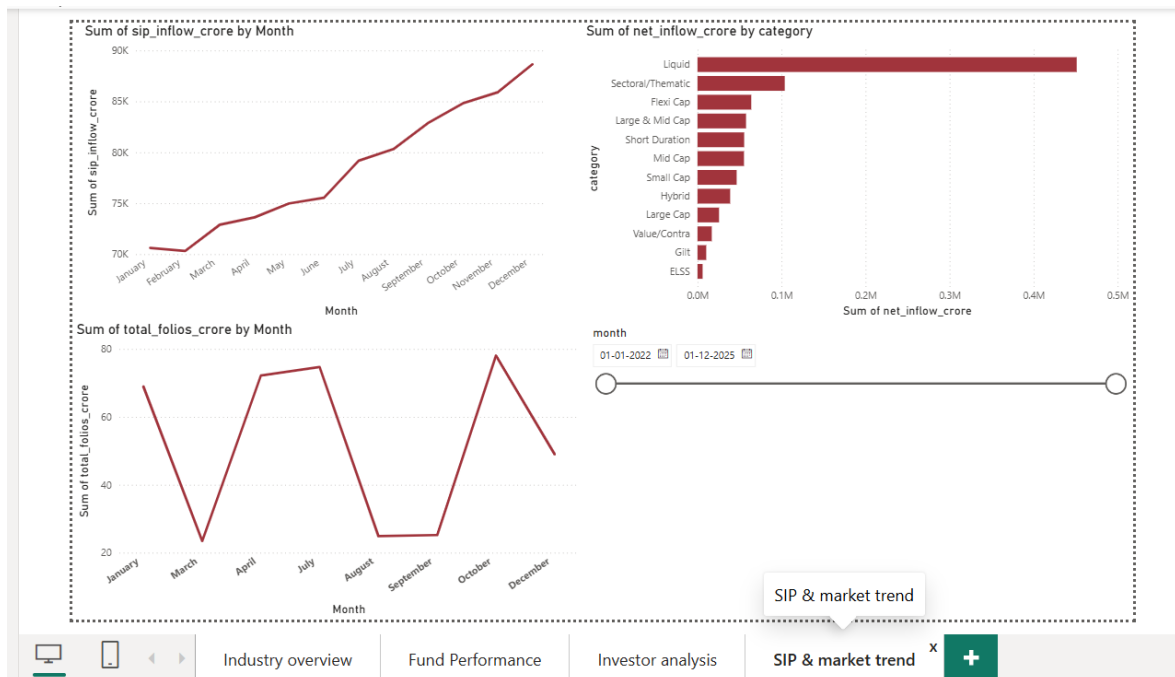
PAGE 2 - Fund Performance: Scatter plot for Risk vs Return. Sortable table with Sharpe, CAGR and Alpha. NAV history line chart.



PAGE 3 - Investor Analytics: Bar chart by state. Donut chart for transaction type split. Column chart by age group.



PAGE 4 - SIP and Market Trends: SIP inflow trend. Category inflows bar chart. Folio growth line chart.



6. Advanced Analytics

VALUE AT RISK: 95% VaR computed for all 40 funds. Small cap funds showed worst VaR of -2% to -3% per day vs large cap at -1% to -1.5%.

ROLLING SHARPE: 90-day rolling Sharpe for top 5 funds. All funds dipped below 1 during 2022 correction but recovered by 2024-2025.

COHORT ANALYSIS: 2024 cohort shows higher average SIP amount than 2022 cohort.

SIP CONTINUITY: Investors with gaps greater than 35 days flagged as At Risk.

FUND RECOMMENDER: Top 3 funds suggested per risk profile using Sharpe ratio and risk grade.

7. Conclusion

1. Indian MF industry is growing rapidly with AUM doubling in 4 years.
2. SIP culture is strengthening with 9.35 crore active SIP accounts.
3. Large cap funds are safe but highly correlated with each other.
4. Small cap funds give higher returns with higher risk.
5. Direct plans significantly outperform Regular plans due to lower expense ratio.

LIMITATIONS: Investor transaction data is simulated. NAV data limited to 40 schemes. Dashboard requires Power BI Desktop.

During this project I found it interesting that SBI Mutual Fund has almost double the AUM of ICICI Prudential. I also noticed that younger investors in the 26-35 age group are investing more through SIP compared to older age groups. Building the ETL pipeline taught me how messy real world data can be and the Power BI dashboard helped me understand how to present data to non technical audiences.